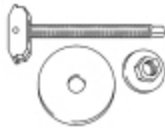


Transmission

 [Ford Ecat Electronic Parts Catalog](#)

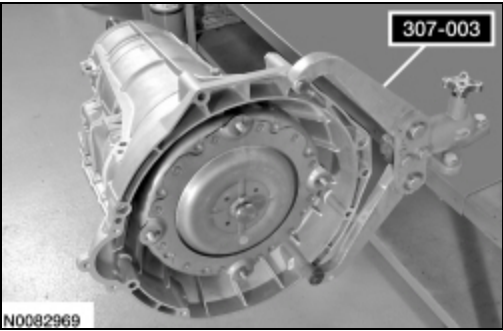
Special Tool(s)

 ST1631-A	Handle, Torque Converter 307-091 (T81P-7902-C)
 ST1186-A	Holding Fixture, Transmission 307-003 (T57L-500-B)
 ST1433-B	Installer, Front Wheel Hub Oil Seal 205-256
 ST2381-A	Remover, Input Shaft Oil Seal 308-375
 E173280	Remover, Koastal Sleeve 307-717
 ST2932-A	Remover, Needle Bearing 307-562 (includes 307-562/1 and 307-562/2)
 ST1282-A	Remover, Pilot Bearing 308-001 (T58L-101-B)

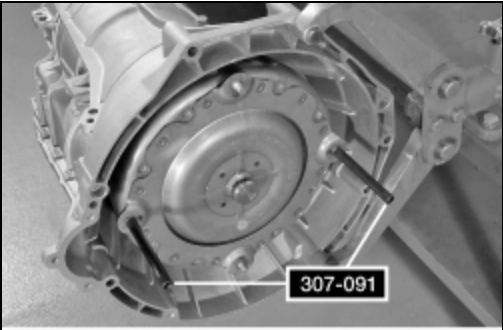
 ST2891-A	Remover, Transmission Fluid Pump 307-553
 ST1104-B	Retaining Ring Pliers 307-343 (T95P-77001-AHR)
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)

All vehicles

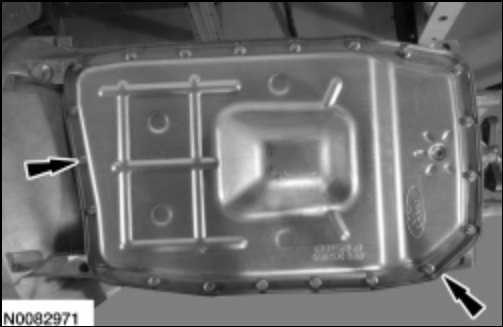
1. Install the Transmission Holding Fixture on the torque converter housing and install the transmission on a bench.



2. Using the Torque Converter Handle, remove the torque converter.



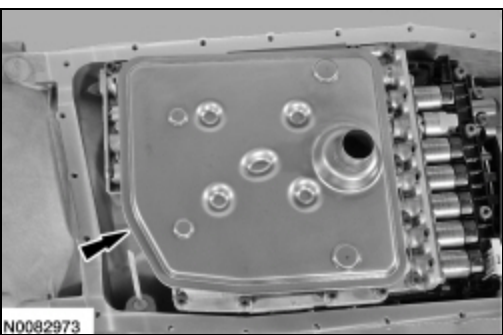
3. Remove the transmission fluid pan bolts and the transmission fluid pan.



4. Remove the transmission fluid pan gasket.



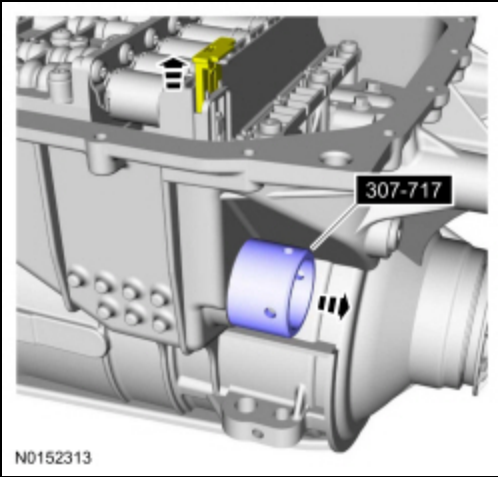
5. Remove and discard the transmission fluid filter.



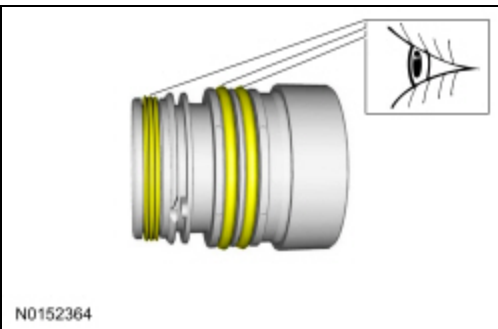
6. Press the release tab and lift up on the bulkhead connector sleeve retainer to release the bulkhead connector sleeve shell.



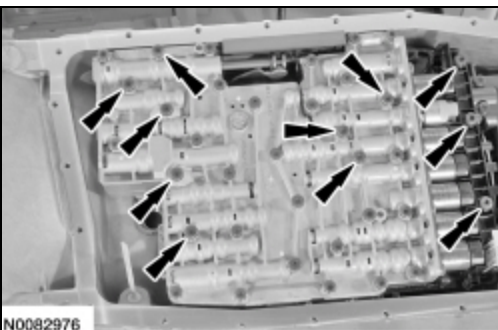
7. With the transmission bulkhead connector sleeve retainer released, use the Koastal Sleeve Remover to pull the outer shell of the bulkhead connector sleeve out of the transmission.



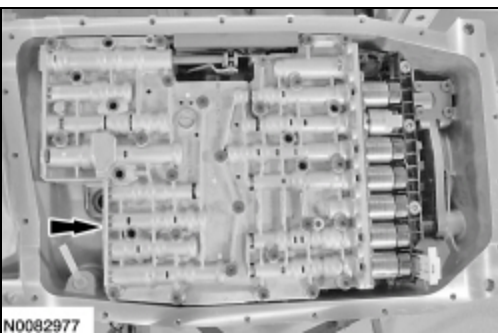
8. Inspect the bulkhead connector sleeve seals.
 - If necessary, replace the seals.



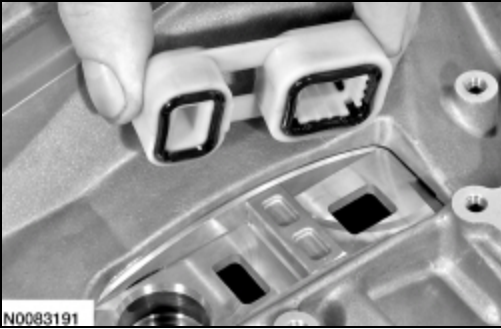
9. Remove the 11 bolts from the main control assembly.



10. Remove the main control assembly.



11. Remove the front pump adapter seal and the transmission fluid filter seal. Inspect for cracks or damage.

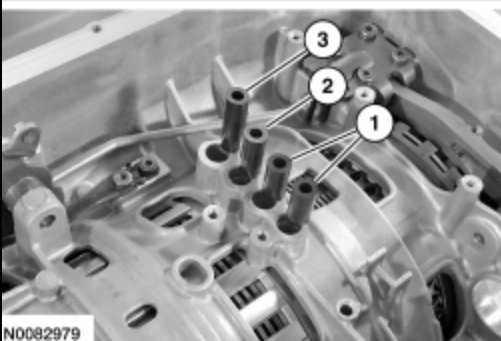
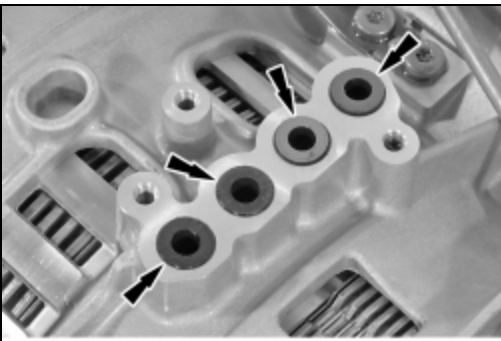


Vehicles equipped with early build transmissions

12. **NOTE:** *Note the color, size and location while removing for correct assembly.*

Remove and discard the 4 center support feed tube seals.

- 1. Center support feed tube seal (7G199 Black)
- 2. Center support feed tube seal (7G087 Green or Orange)
- 3. Center support feed tube seal (7G084 Blue)

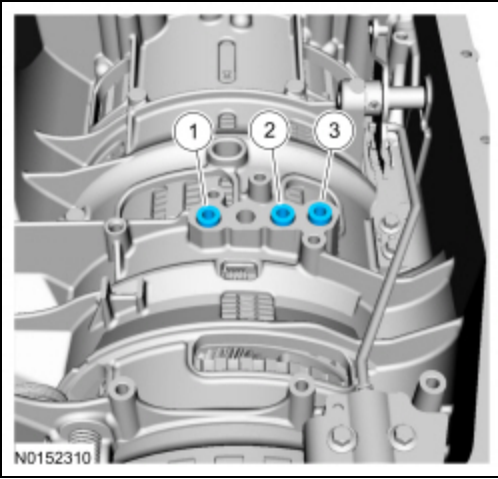


Vehicles equipped with late build transmissions

13. **NOTE:** *Note the color, size and location while removing for correct assembly.*

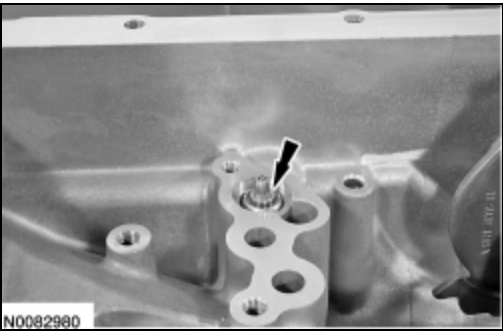
Remove and discard the 3 center support feed tube seals.

- 1. Center support feed tube seal (7G199 Black)
- 2. Center support feed tube seal (7G087 Green or Orange)
- 3. Center support feed tube seal (7G084 Blue)

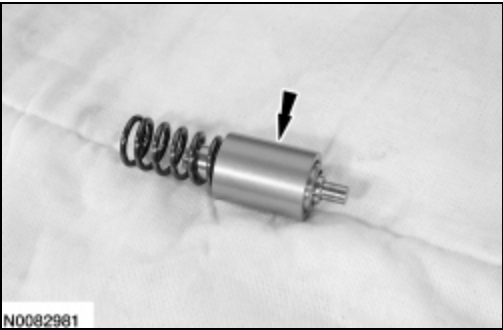


All vehicles

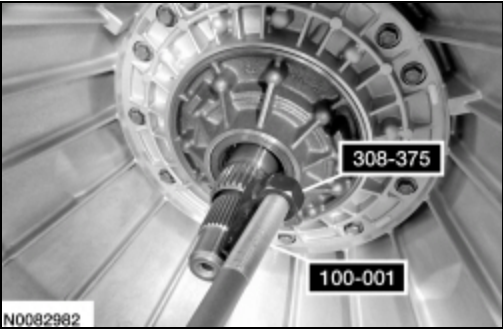
14. Remove the thermal bypass valve from the case.



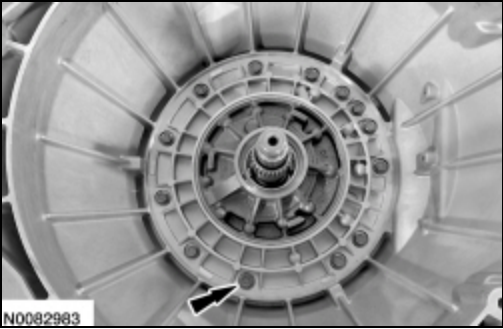
15. Inspect the thermal bypass valve for damage and install a new valve if damaged.



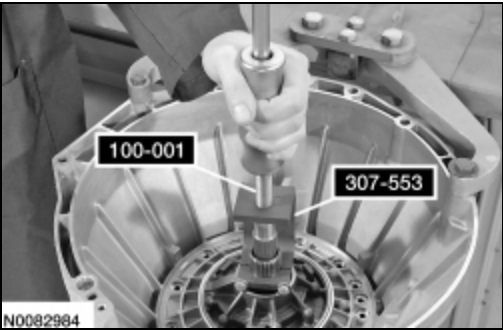
16. Using the Input Shaft Oil Seal Remover and Slide Hammer, remove and discard the front pump seal.



17. Remove and discard the 13 front pump-to-case bolts and washers.

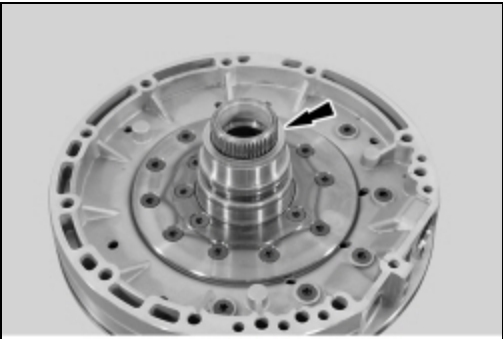


18. Using the Transmission Fluid Pump Remover and Slide Hammer, remove the front pump support.

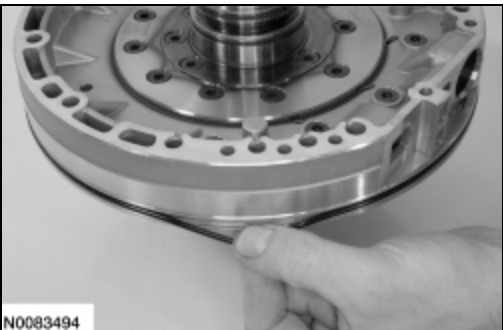


19. **NOTE:** *The selective shim may come out while removing the front pump.*

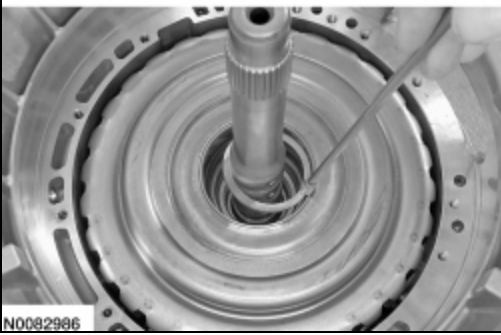
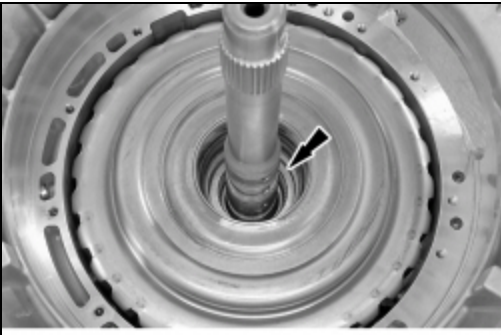
Inspect the back of the front pump for the selective shim and remove If necessary.



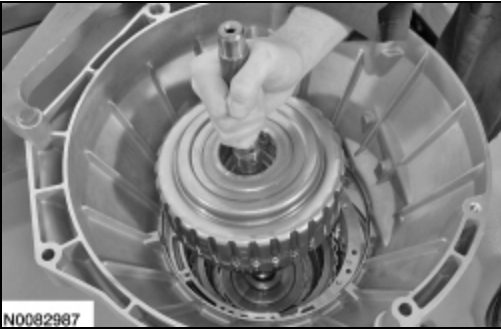
20. Remove and discard the front pump O-ring seal.



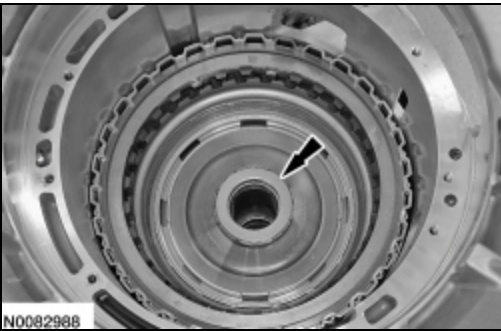
21. If the selective shim is not on the pump, using a magnet, remove it from the forward (A)/overdrive (E) clutch assembly.



22. Remove the forward (A)/overdrive (E) clutch assembly.



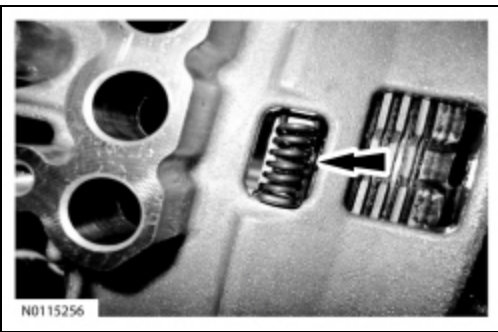
23. Remove the caged T5 roller bearing from the direct (B) clutch assembly.



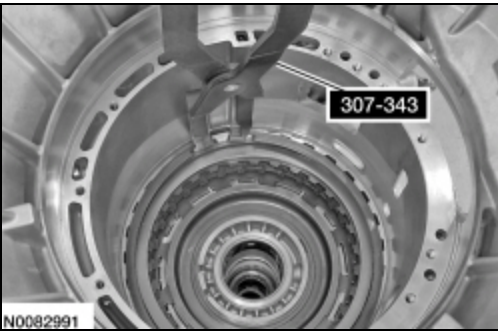
24. Remove the direct (B) clutch assembly.



25. Remove the bias spring.



26. Using the Retaining Ring Pliers, remove the center support snap ring.

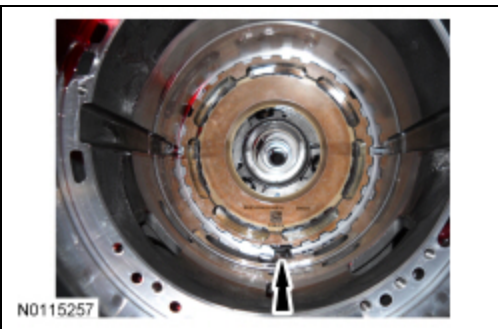


27. Remove the center support and T6 bearing.



28. **NOTE:** When removing the Q.W.C , note the position for assembly.

Remove the Q.W.C .



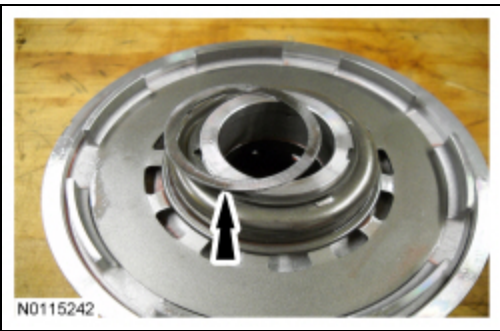
29. **NOTE:** The selective shim might stick to the T7 thrust bearing during removal of the center support.

Remove the selective shim and the T7 thrust bearing.



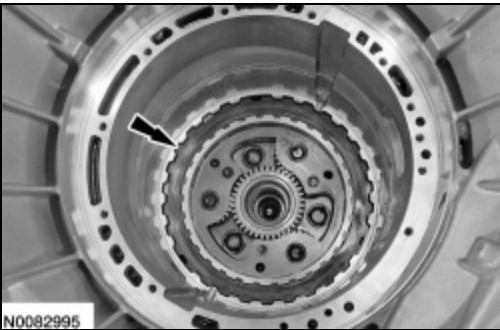
30. **NOTE:** *The selective shim might stick to the back of the center support assembly during removal.*

Remove the selective shim from the center support assembly.

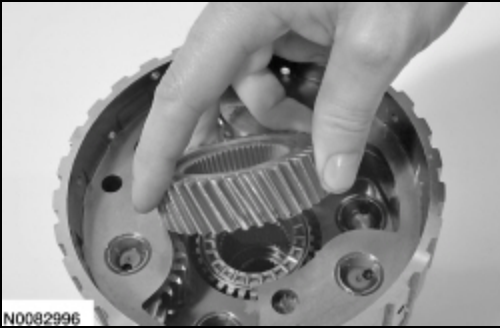


31. **NOTE:** *Some of the clutch plates may stay in the case when removing the planetary carrier and the low/reverse clutch pack. Remove any clutch plates from the transmission that were not removed with the carrier and keep the clutch plates together.*

Remove the planetary carrier and the low/reverse clutch pack as an assembly.



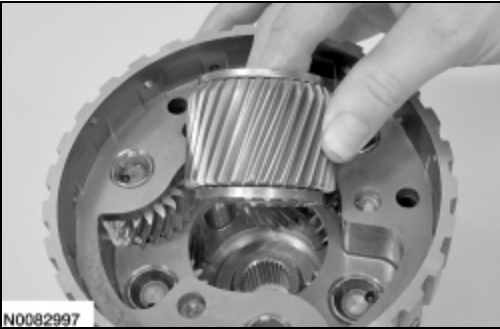
32. Remove the No. 2 sun gear and race.



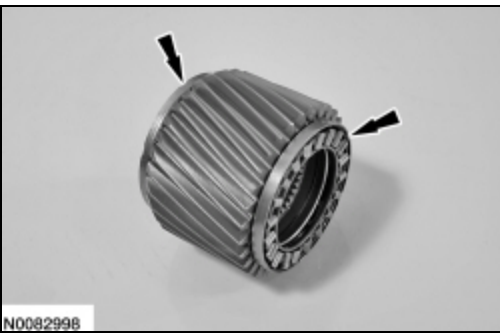
33. **NOTE:** Remove the bearings with the sun gear.

NOTE: When removing the sun gear, note that the tapered edge is facing up toward the torque converter housing.

Remove the No. 3 sun gear from the planetary assembly.



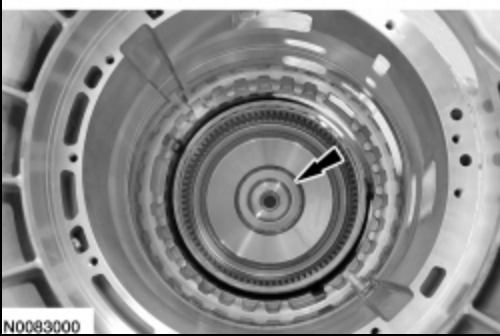
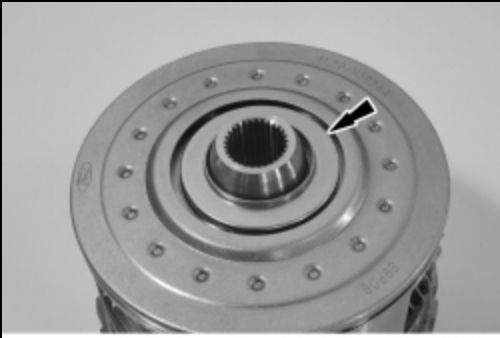
34. Remove the top T8 and the bottom T9 roller bearings from the sun gear.



35. Remove the T9 roller bearing race from the bottom of the carrier.



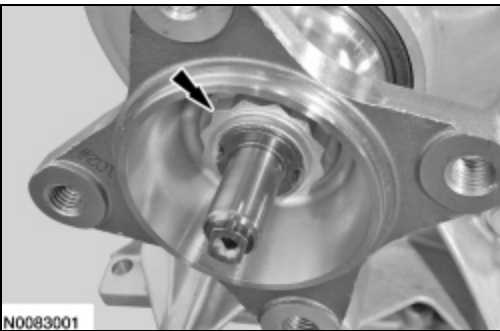
36. Remove the T10 thrust bearing from either the rear planetary carrier or the output shaft assembly.



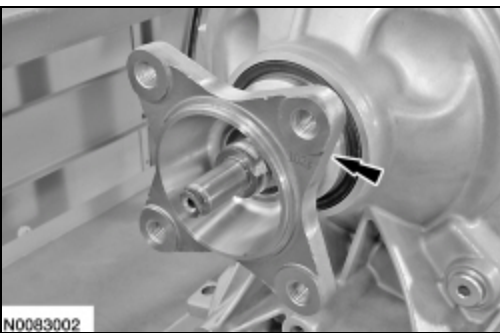
Rear Wheel Drive (RWD) vehicles

37. NOTICE: The output shaft flange retaining nut has been staked to prevent it from coming loose. Prior to removing the nut, remove the stake to prevent damage to the output shaft.

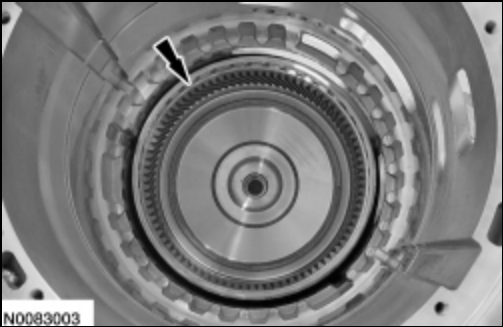
Place the manual control lever in the PARK position and remove and discard the output shaft flange retaining nut.



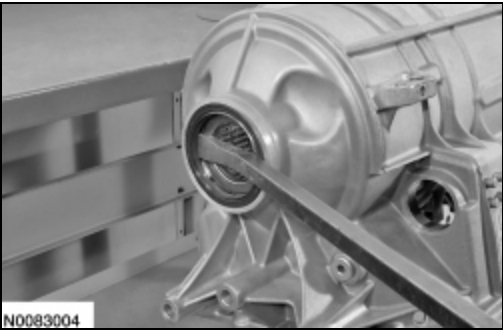
38. Remove the extension housing flange seal and the output shaft flange.



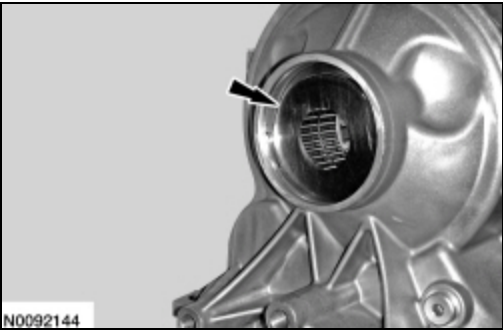
39. Remove the planetary carrier hub, output shaft assembly and T11 bearing.



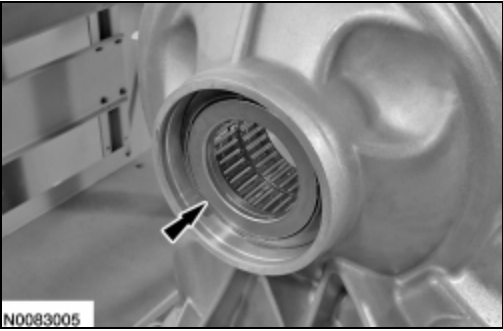
40. Remove and discard the output shaft seal.



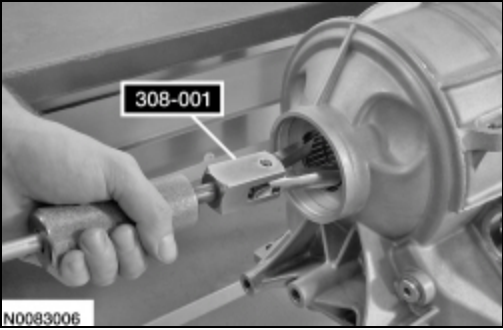
41. Remove the slip plane washer.



42. Remove the T12 thrust bearing.



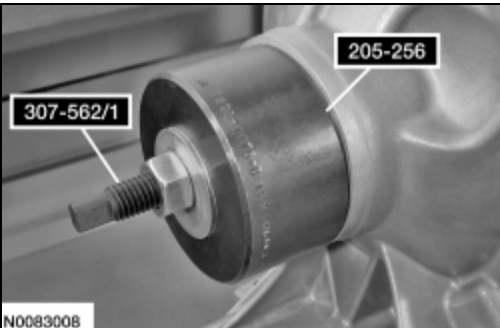
43. Using the Pilot Bearing Remover, remove the thrust bearing spacer.



44. Install the Needle Bearing Remover on the bearing.

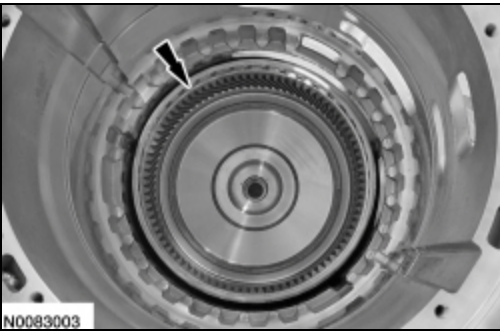


45. Install the Front Wheel Hub Oil Seal Installer on the Needle Bearing Remover and remove the bearing.

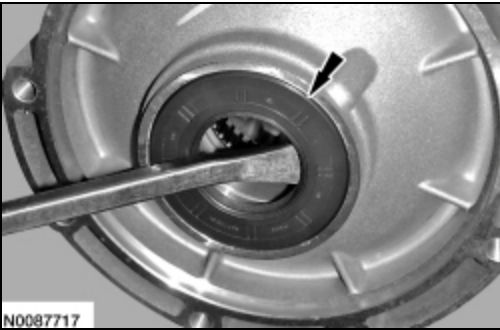


Four-Wheel Drive (4WD) vehicles

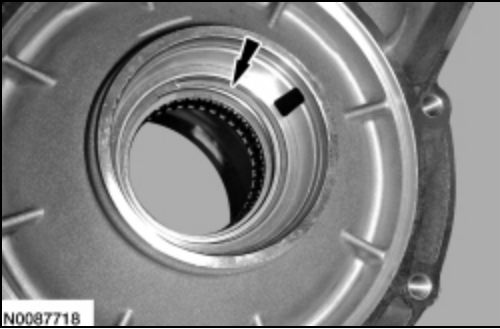
46. Remove the planet carrier hub, output shaft assembly and T11 bearing.



47. Remove and discard the output shaft seal.

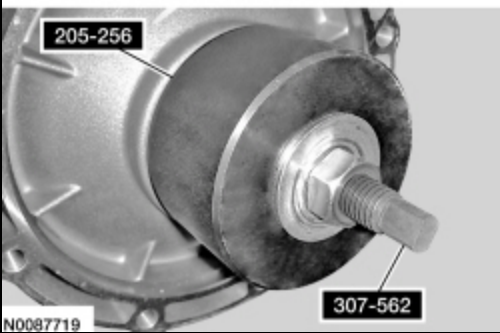
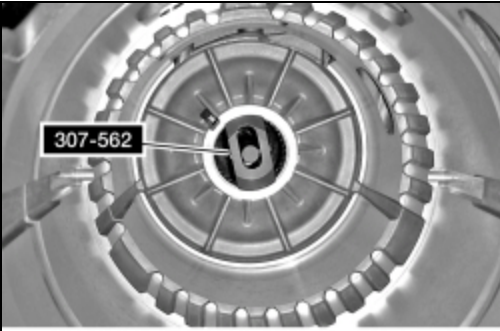


48. Remove the snap ring from the case.



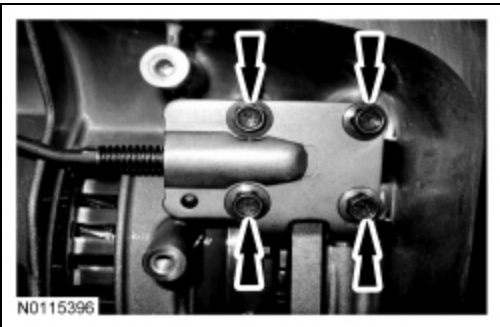
49. **NOTICE:** Use care when installing the Needle Bearing Remover. Look inside the transmission case to make sure that the Needle Bearing Remover contacts the bearing only and not the lip of the case, or damage to the case will occur.

Using the Needle Bearing Remover and the Front Wheel Hub Oil Seal Installer, remove the bearing assembly.

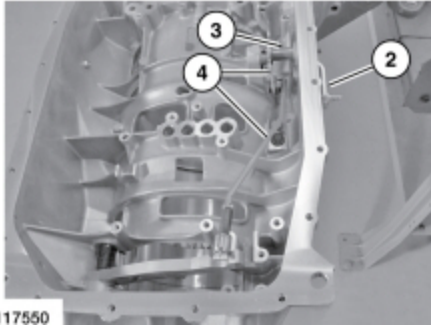
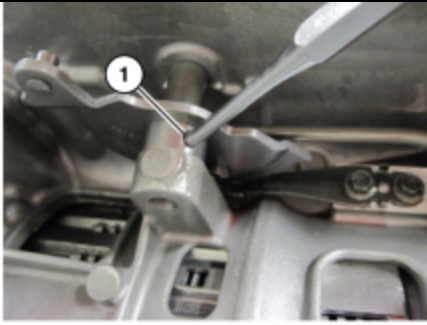


All vehicles

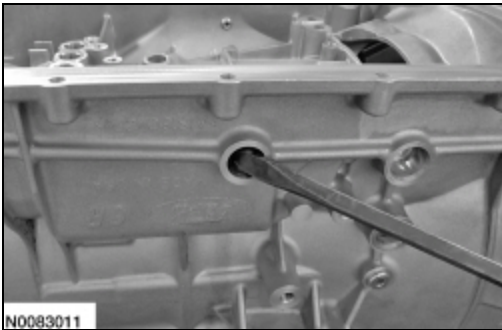
50. Remove and discard the 4 park rod actuating plate bolts and the park rod actuating plate.



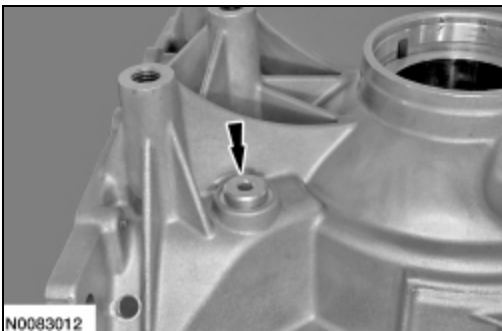
51. Remove the manual control lever detent plate and park pawl actuator rod from the transmission case.
1. Remove and discard the roll pin.
 2. Slide the manual control lever shaft out of the case.
 3. Hold the manual control lever spacer to keep it from falling into the case.
 4. Remove the manual valve detent lever and park pawl actuator rod as an assembly.



52. Remove and discard the manual control lever shaft seal.

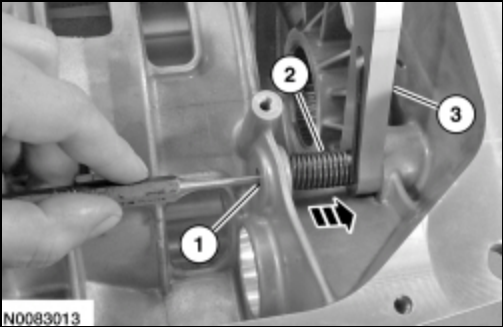


53. Remove the bolt from the back of the case to gain access to the park pawl pin.



54. From the inside of the case, push the park pawl pin out toward the back side of the case while holding the park pawl return spring and the park pawl. Remove the park pawl pin then remove the park pawl and spring.

1. Park pawl pin
2. Park pawl return spring
3. Park pawl



55. Remove and discard the 2 bolts and the park detent spring.

